

RAVI BANDARU

Senior Full Stack Engineer • Angular & React Specialist • AI Driven Engineer

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PROFESSIONAL SUMMARY

AI-driven Senior Software Engineer with 8+ years of expertise developing and delivering enterprise-grade Angular and React applications across healthcare, HR-tech, IoT, and EV sectors. Proven ability to lead frontend initiatives, engineer performant SPAs, and integrate AI-powered capabilities including LLM-driven chatbots, RAG pipelines using ChromaDB and LangChain, and agentic workflows. Combines deep frontend mastery — Angular standalone components, RxJS, NgRx, Performance Optimization Technics, React (Vite + TypeScript) — with full-stack proficiency in Node.js, Supabase PostgreSQL, SQL, and RESTful API design. Sole architect of a production AI chatbot platform (FastAPI · LiteLLM · LangChain · ChromaDB · React) with Minimax AI game engine integration, demonstrating applied AI engineering depth rare among frontend specialists. Adept at translating complex business requirements into scalable, maintainable architectures while mentoring teams, driving Agile delivery, and leveraging cutting-edge AI tooling (OpenAI, LiteLLM, MCP, N8N) to accelerate product development and deliver measurable business outcomes.

CORE COMPETENCIES & TECHNICAL SKILLS

Frontend	Angular (2–18) · React.js · TypeScript · JavaScript (ES2022) · HTML5 · CSS3 · SCSS · RxJS · NgRx
UI Frameworks	Angular Material · PrimeNG · PrimeReact · Syncfusion · Kendo UI · Tailwind CSS · Bootstrap · Ngx-Bootstrap
Backend & APIs	Node.js · Express.js · FastAPI (Python) · RESTful APIs · API Contract Design · SQLAlchemy · Alembic · Supabase · PostgreSQL · SQL
AI & Automation	OpenAI APIs · LangChain · LiteLLM · RAG Pipelines · ChromaDB · Vector Embeddings · LLM Integration · MCP · N8N Automation · GitHub Copilot · Prompt Engineering · Conversation Memory
Testing & Quality	Jest · Karma · Jasmine · Unit Testing · Code Reviews · Debugging
DevOps & Tools	Git · Jenkins · CI/CD Pipelines · Jira · Agile/SCRUM · Webpack · Lazy Loading · Code Splitting · AWS (S3 Storage, EC2 Application Deployments)
Integrations	Stripe Payments · Google APIs · OpenTok · PubNub · SuperSet · Apache Superset, AM Charts
Soft Skills	Frontend Architecture · Technical Leadership · Cross-functional Collaboration · Mentoring · Problem Solving

PROFESSIONAL EXPERIENCE

Senior Software Engineer — Frontend & Full Stack | **Amzur Infotech Pvt Ltd** | Visakhapatnam, India • March 2019 – Present

Angular & React Application Development

- ▶ Developed and delivered 5+ enterprise SPAs using Angular (v8–v18) and React, serving thousands of concurrent users with sub-2-second load times through advanced lazy loading, code splitting, and tree-shaking strategies.
- ▶ Engineered a modular, reusable component library leveraging Angular Standalone Components, reducing new feature development time by an estimated 30% across projects.
- ▶ Designed and implemented role-based access control (RBAC) systems and permissions-driven UI for enterprise-grade applications, ensuring compliance with security and business logic requirements.

- ▶ Optimized Angular change detection strategies (OnPush, Zone.js tuning) and implemented virtual scrolling, reducing rendering bottlenecks by up to 40% in data-heavy dashboards.
- ▶ Delivered fully responsive, cross-browser-compatible interfaces across Angular Material, PrimeNG, Syncfusion, Kendo UI, and Tailwind CSS — meeting WCAG accessibility standards.

API Integration & Full Stack Contributions

- ▶ Collaborated with backend engineers to design RESTful API contracts and contributed to end-to-end full-stack feature delivery using Node.js and Express.js.
- ▶ Integrated real-time data pipelines and dynamic dashboard experiences by consuming complex REST APIs, enabling live data visualization for business stakeholders.
- ▶ Implemented Stripe payment gateway and Google API integrations, enabling commerce and geolocation capabilities in production applications.

AI Engineering & Intelligent Feature Development

- ▶ Built and deployed production-ready AI-powered chatbot interfaces integrated with OpenAI APIs, including dynamic response handling, context-aware conversation flows, and intelligent UI design patterns.
- ▶ Developed LLM-powered features using LangChain, RAG (Retrieval-Augmented Generation) pipelines, and MCP (Model Context Protocol), enabling knowledge-grounded AI responses within web applications.
- ▶ Implemented N8N automation workflows to orchestrate multi-step AI agent pipelines, reducing manual data processing effort across internal operations.
- ▶ Applied advanced prompt engineering techniques and leveraged GitHub Copilot to accelerate feature development, improve code quality, and reduce boilerplate code generation time by ~25%.

Engineering Excellence & Team Collaboration

- ▶ Maintained >85% unit test coverage across modules using Jest, Karma, and Jasmine — enforcing quality gates in CI/CD pipelines via Jenkins and GitHub Actions.
- ▶ Actively participated in Agile/SCRUM ceremonies including sprint planning, backlog grooming, retrospectives, and daily stand-ups — consistently delivering features on or ahead of schedule.
- ▶ Mentored junior developers in Angular best practices, component architecture, and RxJS patterns, fostering a culture of clean, maintainable code.
- ▶ Managed source control, branching strategies, and release workflows using Git and Jenkins, supporting continuous deployment across multiple environments.

Junior IT Consultant | Primor Consultancy Services Pvt Ltd | Visakhapatnam • July 2018 – March 2019

- ▶ Delivered frontend development support on Angular-based client projects, gaining foundational expertise in component-driven architecture, API integration, and Agile workflows.

Junior IT Consultant | Thrill Health and Wellness Pvt Ltd | Visakhapatnam • November 2017 – July 2018

- ▶ Contributed to Angular web development in the health-tech domain, assisting in building responsive interfaces and consuming healthcare APIs in a fast-paced startup environment.

FEATURED PROJECTS

Wellgistics Health ·

Role: Front-End Lead / Angular Developer | **Team:** 15 Engineers | **Stack:** Angular · C# · PrimeNG · Syncfusion · HTML Email Builder

URL: wellgisticshealth.com

- ▶ Led frontend development on a healthcare technology and pharmaceutical logistics platform simplifying prescription fulfillment from manufacturers to patients across specialty pharmacies.
- ▶ Developed an integrated prescription ecosystem dashboard using Angular with Syncfusion and PrimeNG components, enabling real-time fulfillment tracking and pharmacy management workflows.
- ▶ Owned front-end leadership responsibilities — conducting code reviews, defining component architecture standards, and coordinating with a cross-functional team of 6 engineers.

eTeki — On-Demand Technical Interviewing Platform ·

Role: Full-Stack Developer | Team: 16 Engineers | **Stack:** AngularJS · Angular (V8 -V11) · Node.js · ExpressJS · OpenTok · PubNub · CodeEditor · Whiteboard · Screen Sharing

URL: eteki.com

- ▶ Developed a real-time technical interviewing platform enabling live candidate assessment via OpenTok (video), shared whiteboard, collaborative code editor, and screen sharing capabilities.
- ▶ Migrated core modules from AngularJS to Angular 8, improving performance and maintainability while preserving backward compatibility during the transition period.
- ▶ Integrated PubNub for real-time communication channels, enabling low-latency collaboration features critical to the live interview experience.

Evoke Driver Portal — EV Charging Station Locator ·

Role: Full-Stack Developer | Team: 3 Engineers | **Stack:** React.js · Node.js · Google APIs · Stripe · Ngx-Bootstrap

URL: evokesystems.com

- ▶ Engineered a responsive web application for locating electric vehicle (EV) charging stations based on real-time availability using Google Maps and Location APIs.
- ▶ Designed and implemented backend REST APIs in Node.js to enable seamless data flow between the frontend and third-party data sources.
- ▶ Integrated Stripe payment processing for session-based charging payments, delivering a complete end-to-end product within a lean 3-engineer team.

TBL Buildings — IAQ (Indoor Air Quality) Platform ·

Role: Frontend React Developer | Team: 2 Engineers | **Stack:** React.js · PrimeReact · AMCharts

URL: portal.tblbuildings.com

- ▶ Developed a responsive React.js web application for the Orion™ platform — a real-time Indoor Air Quality (IAQ) monitoring solution that bridges live environmental sensor data with automation to support healthier, resilient, and sustainable living and workplace environments.
- ▶ Built dynamic, data-rich dashboards using PrimeReact components and AMCharts for real-time visualization of IAQ metrics including CO₂, humidity, temperature, and particulate matter — enabling actionable insights for building managers.
- ▶ Owned full frontend lifecycle in a lean 2-person team — responsible for feature development, deployment, and maintaining a clean GitHub codebase with structured branching and release workflows.

Full-Stack AI Chatbot Platform (Personal AI Project) ·

Role: Solo Architect & Full-Stack Developer | Team: Personal Project | **Stack:** React (Vite + TypeScript) · FastAPI · LangChain · LiteLLM · ChromaDB · Supabase PostgreSQL · SQLAlchemy (Async) · Alembic · Pydantic

GitHub: github.com/bandaru122/Chat-bot

- ▶ Independently architected and built a production-grade AI chatbot platform from scratch — featuring a FastAPI async backend, React (Vite + TypeScript) frontend, LiteLLM proxy for multi-LLM routing, and RAG (Retrieval-Augmented Generation) pipeline using ChromaDB for vector storage and LangChain LCEL chains for intelligent, context-aware responses.
- ▶ Designed a clean layered backend architecture following separation-of-concerns principles: HTTP-only FastAPI routers → service layer (business logic) → AI layer (LangChain chains, RAG ingestion/retrieval, conversation memory) → database layer (async SQLAlchemy ORM + Supabase PostgreSQL with Alembic migrations) — demonstrating enterprise-grade software design.
- ▶ Built a full REST API surface covering note CRUD, LLM chat, note summarization, and Google Sheets integration via service account — validated by 18-check automated setup test suite covering ChromaDB, LiteLLM, DB connection, and credentials.
- ▶ Implemented an AI-powered Tic Tac Toe game engine within the chat interface using a Minimax algorithm with alpha-beta pruning — achieving 95%+ AI win rate on Hard difficulty, with full TypeScript React component architecture, difficulty levels, and intent-detection integration into the chat flow.
- ▶ Demonstrated advanced prompt engineering, multi-turn conversation memory management, and embeddings routing via LiteLLM proxy — mirroring patterns used in real-world enterprise AI assistant and copilot products.

EDUCATION

Bachelor of Engineering (B.E.) | Raghu Institute of Technology, Visakhapatnam 2012 – 2015 • CGPA: 7.10

Diploma in Engineering | Govt Polytechnic College, Parvathipuram 2009 – 2012 • Score: 80.40%

SSC (10th Grade) | SVD Gangadhar High School, Vizianagaram 2008 – 2009 • GPA: 89.89%

TECHNICAL ACHIEVEMENTS & HIGHLIGHTS

- ▶ **Enterprise Architecture:** Successfully delivered full-lifecycle enterprise solutions over an 8-year tenure, consistently bridging complex business requirements with scalable, maintainable architectures in Angular, React, and AWS ecosystems.
- ▶ **AI Platform Engineering:** Architected and deployed a production-grade AI platform (FastAPI, LangChain, ChromaDB) featuring a custom RAG pipeline and multi-model routing via LiteLLM; delivered an enterprise-ready solution with async ORM architecture and 100% automated test validation.
- ▶ **Technical Leadership:** Served as Angular Tech Lead for Wellgistics Health, guiding a 6-engineer team in defining component standards and architectural patterns; maintained zero critical production incidents for a high-volume healthcare logistics platform.
- ▶ **Engineering Efficiency:** Accelerated development velocity by 25% by pioneering the strategic adoption of AI-assisted workflows, including GitHub Copilot integration, N8N automation pipelines, and multi-provider LLM proxy patterns.
- ▶ **Cross-Domain Problem Solving:** Demonstrated advanced technical depth by integrating a complex algorithmic game engine (Minimax with alpha-beta pruning) into a React frontend, showcasing proficiency in both front-end UI architecture and backend algorithmic complexity.
- ▶ **High-Scale Reliability:** Established rigorous quality gates and CI/CD protocols that sustained >85% unit test coverage across multiple enterprise SPAs, ensuring seamless reliability for thousands of concurrent users.